

ARJUN PRASAD

Senior Backend Engineer | Distributed Systems & Payments

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PROFESSIONAL SUMMARY

Backend engineer with 4.5+ years building high-throughput payment systems at PhonePe (500M+ users). Currently on the UPI Core Transactional team handling 20,000+ RPS and 500M+ daily transactions. Previously owned end-to-end BBPS infrastructure processing 4M+ transactions/day. Built 2 greenfield systems from scratch, created an org-wide async messaging framework reducing infrastructure costs by 45%, led Active-Active DC migration with circular replication, and overhauled a critical pipeline reducing stuck transactions from 20,000/day to <100. Seeking Senior Backend Engineer roles in distributed systems and payments.

TECHNICAL SKILLS

Languages: Java (8-17), SQL, Python
Frameworks: Dropwizard, Spring Boot, Hibernate, JUnit, Mockito
Data: MariaDB (Sharded, Circular Replication), Aerospike, RabbitMQ, MemQ, Event-Driven Architecture
Systems: Microservices, Active-Active DC, Docker, CI/CD, Nginx, System Design, DB Sharding, REST APIs, DR/BCP

PROFESSIONAL EXPERIENCE

PhonePe -- UPI Core Transactional Team

Mar 2026 - Present

SDE-2 (Backend) | 20,000+ peak RPS | 500M+ UPI transactions/day | Pune, India

- Operating on India's highest-traffic payment rail -- handling 20K+ requests/second with mission-critical uptime and zero tolerance for data inconsistency
- Building E2C refund status tracking for 2-2.5 lakh daily refunds -- designing NPCI contract architecture and UDIR dispute resolution flows for cross-PSP integrations (GPay to PhonePe)
- Owning external PSP refund resolution to provide real-time refund visibility where users previously had to check bank statements manually

PhonePe -- Bharat BillPay (BBPS) Pod

Feb 2022 - Feb 2026

Software Engineer to SDE-2 (Backend) | 500 peak RPS | 4M+ billpay transactions/day | Pune, India

UPMS -- Greenfield Unified Presentment Microservice (Built 0 to 1)

- Designed and shipped a new microservice independently: sharded MariaDB, proxy layer, reconciliation engine, biller kill-switches, alerting -- scaled from zero to 500+ RPS in 5 months
- Replaced polling with event-driven push presentment driving 20% transaction growth; completed NPCI sandbox certification; migrated 100% monolith traffic with zero downtime

Axon -- Org-wide Async Messaging Framework

- Created reusable framework abstracting RabbitMQ/MemQ with unified publish API, fallback strategies, dynamic worker pools -- adopted across PhonePe engineering
- Achieved 45% infrastructure cost reduction org-wide; per-service integration time reduced from days to minutes

Active-Active Data Center Migration

- Redesigned DB schema for circular replication across DCs; migrated async workers to MemQ for DC-aware routing -- zero-downtime failover, 50% blast radius reduction

Status Check Pipeline Overhaul

- Re-architected status reconciliation: stuck transactions from 20,000+/day to <100, resolution time 9+ hours to <10 min, eliminated wrong refund risk

Piped Gas Meter Reading (GFF 2025)

- Built camera-based meter reading end-to-end in 1 week (contract to production). Demonstrated at Global Fintech Festival 2025 -- PhonePe-exclusive integration

Additional Contributions

- Paylink migration: +15% monthly transactions, zero downtime | Loan API 1.1: led NPCI spec compliance independently
- Launched 6 bill categories with automated SOP | Mentoring 3 engineers | 117 code reviews | 6 DR drills led
- Driving design decisions, production triage, and cross-team unblocking across the BBPS pod as technical anchor

Samsung Research Institute

Jul 2021 - Feb 2022

Software Engineer | Noida, India

- Reduced audio device setup time by 30% through Bluetooth module optimization across 1,000+ device SKUs
- Earned Samsung SWC Professional certification in Data Structures & Algorithms (competitive internal exam)

EDUCATION

B.Tech in Computer Science & Engineering

2017 - 2021

National Institute of Technology, Delhi

RECOGNITION & IMPACT

- "Distinguished" performance rating (highest tier at PhonePe) -- "Completed the implementation flawlessly and quickly" -- Engineering Manager
- Scale progression: built system at 500 RPS, now operating at 20,000+ RPS -- consistently entrusted with increasing scope and ownership
- 345+ merge requests merged | 605 JIRA tasks delivered | 356 architecture docs | 4 consecutive strong performance cycles
- Represented PhonePe at Global Fintech Festival 2025 with live demo of Piped Gas Meter Reading integration